

DILIPKUMAR NEHRU

Backend Engineer • Node.js • API & Microservices Specialist

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Professional Summary

Results-driven Backend Engineer with 3+ years of experience building scalable, high-performance server-side systems using Node.js, Express.js, MongoDB, and MySQL. Delivered production APIs handling 50,000+ daily requests for enterprise CRM and AI analytics platforms. Proven track record in asynchronous processing (RabbitMQ), database performance optimisation, and cross-functional Agile delivery. Complementary QA automation background (Selenium, Postman, JMeter) enables end-to-end ownership from development through release.

Career Highlights

Key quantified achievements across 3+ years:

10,000+ Daily API requests handled on NovoNordisk CRM and UrbanPiper CRM	35% Reduction in API latency via MongoDB indexing	40% Faster regression cycles with Selenium automation
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Work Experience

Software Engineer – Backend Developer | **Zuqo Corporation India Pvt Ltd** | BengaluruDec 2023 – May 2025

- Designed and maintained scalable RESTful APIs using Node.js and Express.js, serving 50,000+ daily requests for enterprise web applications with 99.5% uptime.
- Reduced MongoDB query response times by 35% through strategic indexing, query optimisation, and connection pooling across the NovoNordisk CRM platform.
- Implemented asynchronous background task processing using RabbitMQ, cutting average job completion time from 8s to under 2s for UrbanPiper AI Analytics.
- Integrated MongoDB and MySQL databases with backend microservices, standardising schema design patterns that reduced inter-service data inconsistencies by 60%.
- Collaborated with 6-member cross-functional team in 2-week Agile sprints, consistently delivering features on schedule with zero critical production rollbacks.
- Debugged and resolved 30+ production incidents, improving application stability and reducing Mean Time to Resolution (MTTR) by approximately 45%.
- Enforced clean code standards, Git branching strategies (GitFlow), and code review practices, reducing post-merge defect rate by 25%.
- Implemented JWT-based authentication and OAuth 2.0 authorisation flows across multiple APIs, eliminating session vulnerabilities and enabling secure third-party integrations.
- Integrated Socket.io to enable real-time WebSocket event notifications and live dashboard updates, reducing client-side polling overhead by 80%.

Software Test Engineer – QA Automation | **Zuqo Corporation India Pvt Ltd** | BengaluruApr 2022 – Dec 2023

- Built and maintained a Selenium automation test suite covering 200+ functional and regression test cases, cutting manual testing effort by 40%.
- Performed comprehensive API testing and backend validation using Postman and JMeter, identifying 80+ defects before production deployment.
- Created structured bug reports, test plans, and troubleshooting runbooks adopted as the QA documentation standard across the team.

- ▶ Collaborated directly with backend developers to triage and resolve defects, reducing average bug resolution cycle from 5 days to 2 days.

Project Highlights

NovoNordisk CRM — Healthcare Enterprise Platform

- ▶ Architected and delivered scalable backend APIs (Node.js, Express.js, MongoDB) handling 50,000+ daily requests for a global pharmaceutical CRM system.
- ▶ Implemented JWT authentication, OAuth 2.0 flows, field-level encryption, and RBAC for HIPAA-aligned secure data handling of sensitive patient and sales records.
- ▶ Optimised 15 high-frequency MongoDB aggregation queries, reducing dashboard load times by 35% and improving UX for 500+ concurrent users.

UrbanPiper AI Analytics — F&B Analytics Dashboard

- ▶ Built event-driven backend microservices with RabbitMQ and Socket.io WebSockets to process and push real-time analytics data streams to 100+ restaurant client dashboards.
- ▶ Reduced asynchronous job processing latency from 8s to under 2s by redesigning queue consumers and introducing dead-letter queue error handling.
- ▶ Delivered REST APIs consumed by React analytics dashboards, enabling real-time KPI monitoring with sub-500ms response times.

Client Support & Backend Maintenance

- ▶ Provided Tier-2 backend production support for 5+ client applications, diagnosing and resolving scalability bottlenecks under SLA constraints.
- ▶ Refactored legacy API endpoints, reducing average response time by 20% and eliminating 3 recurring timeout errors in peak-traffic scenarios.

Technical Skills

Languages	JavaScript (ES6+), Node.js, TypeScript (basic), SQL, Python, Java
Frameworks	Express.js, Socket.io
Databases	MongoDB, SQL, PostgreSQL
Messaging	RabbitMQ, Socket.io (WebSockets)
Auth & Security	JWT (JSON Web Tokens), OAuth 2.0, MD5, HTTPS/TLS, bcrypt
Backend	RESTful APIs, WebSockets, Microservices, API Integration, Async Programming
Testing	Postman, Selenium WebDriver, JMeter, API Automation, Manual Testing
DevOps / Tools	Git, Bitbucket, GitHub, Linux CLI(working knowledge)
Methodologies	Agile Methodology, CI/CD awareness

Certifications

- ▶ Selenium Automation & Manual Testing — QSpiders, Bangalore (2022)
- ▶ MongoDB Basics (M001) — MongoDB University [Recommended Next]
- ▶ Node.js Application Developer — OpenJS Foundation [Recommended next]

Education

Bachelor of Engineering – Electronics & Communication Engineering (ECE) *Graduation: 2020*
RVS Technical Campus, Coimbatore (Anna University)

Diploma – Electronics & Communication Engineering (ECE) *2012 – 2015*
Krishnasamy Memorial Polytechnic College, Cuddalore